Proceedings of the Fifth International Conference on Trends in Electronics and Informatics (ICOEI).
IEEE Xplore Part Number:CFP21J32-ART; ISBN:978-1-6654-1571-2

Performance Evaluation of Distributed Machine Learning for Cardiovascular Disease Prediction in Spark

Yogesh Kumar Gupta[1]
[1]*Assistant Professor, Department of Computer Science, Banasthali Vidyapith, gyogesh@banasthali.in*

Surbhi Kumari [2]
[2]*M.Tech(CSE) Student, Department of Computer Science, Banasthali Vidyapith,* surbhiroy25july@gmail.com

***Abstract*— On a daily basis, human life is suffering from different types of diseases that's why their life is in distress. Cardiovascular disease is a generic category of disease that is effective in spreading infections and notably, it affects the heart and veins. It is observed that cardiovascular diseases are becoming modest in elderly people besides in young people too. It is very requisite to foreshow this kind of disease in the beginning phases; many types of tests are used for diagnosticating these ailments. This implementation has been done by using a big data tool that is Apache Spark and using spark's MLlib and PySpark libraries which are integrated with it. Apache Spark is among the most widely used big data technologies, and it is a stack of some libraries that are Spark SQL, Spark MLlib, Spark Streaming, etc. This research work aims to build a prediction model to predict whether individuals have cardiovascular disease or not, using machine learning classification techniques which include logistic regression, decision tree, support vector machine, random forest, and gradient-boosting tree classifier and also applied hyperparameter tuning and cross-validation with 5-fold to improve the performance of models. They compared the evaluation of all applied machine learning models and the results observed that the Gradient-Boosting Tree Classifier achieved better Accuracy (73.20%) and Area Under ROC value (0.8002).**

Keywords— Cardiovascular Disease; Distributed Machine Learning; Classification; Logistic Regression; Decision Tree; Support Vector Machine; Random Forest; Gradient-Boosting Tree; Apache Spark

I. INTRODUCTION

Cardio Vascular Disease is the most renowned hazardous infection all over the world. Every year, approximately 17 million individuals die reason of cardiovascular ailment that is stated by the World Health Organization. Therefore, we need a framework that can predict cardiovascular ailments and preventing the serious problems of this disease in the future, and fix them before augmentation. Heart failure is a cardiovascular disease, which can occur due to many reasons such as coronary artery disease, hypertension, cardiomyopathies, vasculitis, and others. It can influence the ability of the heart to pump properly. In this research work, we have designed a classification model to predict cardiovascular disease using PySpark and MLlib (machine learning library) which are unified with Apache Spark. We have used various classification techniques (machine learning algorithms) for the prediction model that concluded logistic regression classifier (LR), decision tree classifier (DT), support vector machine classifier (SVM), random forest classifier (RF), and gradient-booting tree classifier (GBT). Furthermore, used diverse performance methods such as Roc Auc score, Accuracy, Precision, and Recall to enumerate. Apache Spark is a free and open-source distributed platform that provides fault tolerance, flexibility, and reusability, etc. Resilient distributed Datasets is an abstraction that is provided by the spark to endorse these applications adroitly. And it can be held in memory between queries, without the need for duplication. Rather, they use lineage to recover lost data in the event of a failure. There are the key findings of this paper: Using Spark machine learning library and big data system: Apache Spark, creating a framework that can predict cardiovascular disease. Creating a framework that can examine cardiovascular disease data and predict whether an individual has cardiovascular disease or not.

II. LITERATURE REVIEW

In 2020, Komal Kumar et. al [7] developed a proposed system for heart disease. They have used a heart disease dataset that was gathered from the UC Irvine(UCI) Repository, which had 10 attributes. The classification techniques random forest, decision tree, logistic regression, k-nearest neighbors, and support vector machine had used for building a prediction model of cardiovascular ailment. After comparing the performance of models, the researchers have shown that the random forest machine learning model obtained the best accuracy with 85.71% and Area Under ROC of 0.8675. Therefore, they used a random forest machine learning model that aids in building a system to classify the patients who are affected with Cardiovascular sickness.

In 2015, Dewan, Ankita Sharma, Meghna et al. [8] proposed a hybrid technique with the ability to solve complex skepticism which is indispensable for the prognosis of cardiac disease which may aid doctors to diagnose the condition. The proposed hybrid technique is based on a dataset with 13 attributes that were taken from the UCI repository. The evaluation matrix used is Accuracy and sensitivity. Comparisons perform among decision trees, Naïve Bayes, SVM & ANN. Results show that ANN outperforms all others classifications for non-linear data.

In 2019, Shamsollahi, M Badiee, A Ghazanfari et al. [9] built a model using an amalgam of descriptive and predictive

2021 5th International Conference on Trends in Electronics and Informatics (ICOEI) | 978-1-6654-1571-2/21/$31.00 ©2021 IEEE | DOI: 10.1109/ICOEI51242.2021.9452955

Authorized licensed use limited to: University of Saskatchewan. Downloaded on August 10,2021 at 02:37:47 UTC from IEEE Xplore. Restrictions apply.

analytics of KDD (Knowledge Discovery in Databases). The authors are determined the number of clusters using clustering indices. After that, the authors have applied some decision tree methods and artificial neural networks for all clusters. They used the original dataset for building a model that is gathered from a heart clinic database. The results have shown that the CART decision tree model attained the best of all methods.

In 2019, Makumba, Dominic Obwogi Cheruiyot, Wilson Ogada, Kennedy et al. [10] developed a model using data mining on heart disease prognosis that can support the making a decision. There are used such decision trees, naive bayes, k-nearest neighbors, and also Waikato Environment for Knowledge Analysis application programming interface. Dataset for the proposed model has been accessed from UCI with 13 attributes. The confusion matrix has been used for the evaluation of the models. The researchers concluded that the proposed framework provides a better prediction to prognostic of the coronary artery disease from the present-day blueprint.

In 2010, Khosla et al. [11] have implemented the Cox proportional hazards model using data mining methods for the prediction of stroke diseases. They performed implementation using the CHS (Cardiovascular Health Study) dataset. They concluded that the support vector machine classifier attained the highest evaluation using the ROC curve.

In 2020, Muktevi Srivenkatesh [12] developed a prediction model for cardiovascular disease and the dataset is taken from Kaggle.com for creating a proposed model. There are used various classification models that are consist of logistic regression classifier, support vector machine classifier, random forest classifier, and naive bayes classifier. They have using metrics accuracy, sensitivity, and specificity to attain the evaluation of the diagnosis model. After evaluating and comparing the models, the authors obtained a better accuracy with 77.06 % of the logistic regression model.

In 2020, Kaur, H. et al. [13] developed a predictive model that uses different machine learning algorithms which are KNN, Linear SVM algorithm, RBF-kernel SVM algorithm, and ANN algorithm. Pima Indian diabetes dataset has been trained and validated for predicting diabetic and nondiabetic patients. The results have shown the Linear SVM model accomplished a good accuracy with 89% out of all models.

In 2020, Ahmed et al. [14] created a system using the Spark ML library which is accompanied by the big data platform Apache Spark. To build the prediction model for diabetes patients, they have used classification methods that included are support vector machine algorithm, decision tree algorithm, logistic regression algorithm, naive bayes algorithm, and random forest algorithm. Following that, evaluated all models using some matrices such as accuracy, recall, and precision then they found that the logistic regression model accomplished the best percentage score of Accuracy (82%), Recall (92%), and Precision (82%).

In previous studies, authors who worked with small cardiovascular datasets and evolved models have only used conventional methods of machine learning algorithms that can use to predict Cardiovascular disease. In this paper, we have worked with a large dataset and developed a predictive model using PySpark and Spark's MLlib library (distributed machine learning) that are united on the Apache Spark platform that can able to detect and predict cardiovascular ailment patients.

III. PROPOSED METHODOLOGY

We worked on a cardiovascular ailment dataset of 70000 cases with 13 attributes has collected from Kaggle. This dataset consists of features which are age, gender, height, weight, systolic blood pressure, diastolic blood pressure, cholesterol, smoke, alco, active and cardiovascular. From the dataset, the feature cardiovascular has two binary values 0 and 1. Dataset 34979 (1) demonstrates that an individual having a Cardiovascular ailment and dataset 35021 (0) demonstrates that an individual no Cardiovascular ailment. In table 1 has been shown the features name and description of the cardiovascular disease dataset. Figure. 2 presents the architecture of the cardiovascular disease prediction framework. In the proposed framework, we started with loading the dataset into the Dataframe. Performed cleaning the dataset using the Pandas tool, and then applied the preprocessing techniques. Following that, the dataset is divided into training (70%) and training (30%) datasets now the machine learning algorithms are applied to the divided (training & testing) dataset. The train-test split technique is used to evaluate the performance of any supervised machine

Table 1: Attributes name and description of cardiovascular ailment dataset learning algorithm. Classification and regression problems are solved using it. There is divide the dataset into two subsets, the first subset which has most of the data is used to fit the model and is known as the training dataset. The second subset which having a smaller portion of the data is used to evaluate the fit model and is known as the testing dataset. At last, we evaluated the classification models using metrics which are Area Under ROC and Accuracy.

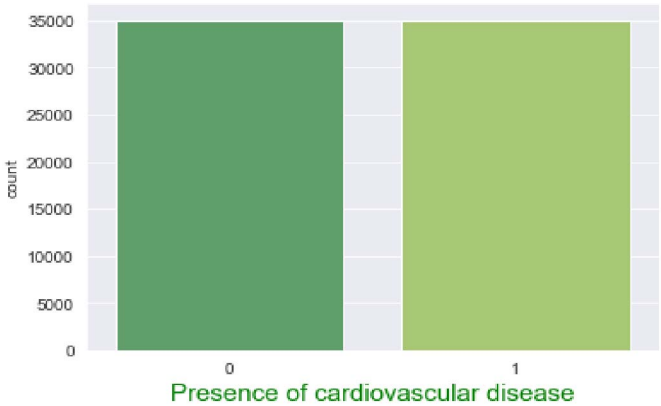


Figure 1. View of the Target Class

Authorized licensed use limited to: University of Saskatchewan. Downloaded on August 10,2021 at 02:37:47 UTC from IEEE Xplore. Restrictions apply.

Data balancing is requisite for consistent results since the data balancing graph shows that both the target classes are Equal. Figure 3 illustrates the target classes, where "1" denotes a patient with cardiovascular disease and "0" represents a patient who does not have cardiovascular disease.

Table 1: Attributes name and description of cardiovascular ailment dataset

Serial Number	Attribute	Remarks
01	ID	ID number
02	Age	in Days
03	Gender	1-women,2-Men
04	Height	In Cent Meter
05	Weight	Kilo Grams
06	Systolic Blood Pressure	Systolic Blood Pressure
07	Diastolic Blood Pressure	Diastolic Blood Pressure
08	Cholesterol	1-Normal,2-Above Normal, 3-Well Above Normal
09	Glue	1-Normal,2-Above Normal, 3-Well Above Normal
10	Smoke	Whether Patient Smokes or Not
11	Alco	Binary Feature
12	Active	Binary Feature
13	Cardiovascular	Target Variable

The proposed framework is explained with all phases in detail below.

A. Data Cleaning

Data cleaning is a process that involves the correction and detection of errors from datasets. It is important because this process gives the highest quality of information that improves model performance. In this work, we removed irrelevant columns and dropped the duplicate rows. Also, we replaced the strange value in columns. The id attribute of each case is detached from the dataset because it is irrelevant to the diagnostic model.

B. Data Pre-processing

Data Preprocessing is an essential phase in machine learning techniques that involves cleaning, standardization, transformation feature extraction & selection, and so on. We get some important insights when removed garbage data which provides better productivity. In our work, we applied feature extraction for creating the new features. After that, we used Vector Assembler to transform the features into a vector, and then applied StandardScaler to the features.

C. Machine Learning Algorithms

Machine learning (ML) is a collection of algorithms that are capable of learning to execute tasks i.e. forecast and classify with help of a dataset. In this phase, for predicting the model of cardiovascular ailment, machine learning algorithms are implemented on splitted (training & testing) dataset. We have used classification algorithms that included logistic regression classifier, decision tree classifier, support vector machine classifier, random forest classifier, and gradient-boosting tree Classifier. Spark MLlib (machine learning library) has been used to apply machine learning models.

- Logistic Regression Algorithm

Logistic regression is a classification algorithm that uses for binary classification. We assume all predictors are independent of each other and the outcome obtained by the dependent variable that comes with two target values which are 0 or 1, or yes or no. The execution of the logistic regression model has been shown in Figure 2.

The computational complexity of the Logistic Regression Algorithm: -

n= number of training examples

d= number of dimensions of the data

Train Time Complexity=O(nd)

Space Complexity = O(d)

- Decision Tree Algorithm

The decision tree is an algorithm that helps to build a classification and regression model. It generates a set of rules which used to categorize the data. It has the shape of a tree and offers a high level of precision and reliability. And it is

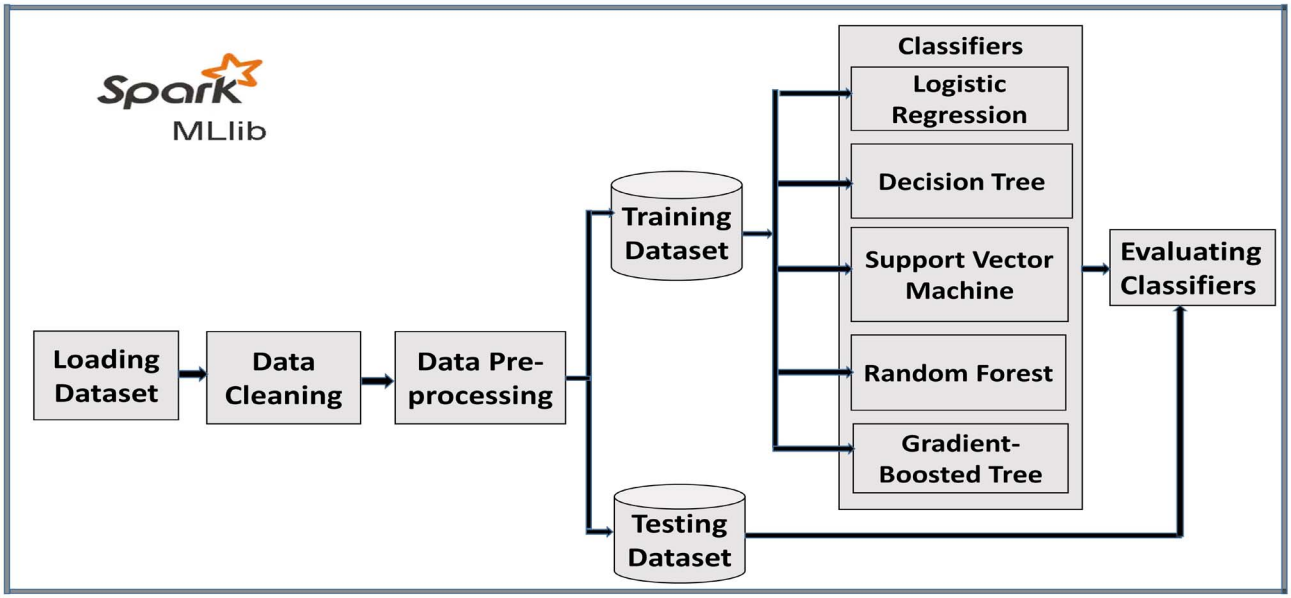


Figure 2. Proposed Framework

Authorized licensed use limited to: University of Saskatchewan. Downloaded on August 10,2021 at 02:37:47 UTC from IEEE Xplore. Restrictions apply.

capable of dealing with both numerical and categorical data. The execution of the decision tree model has been shown in Figure 3. The computational complexity of Decision Tree Algorithm: -
n=number of points in the Training set
d=dimensionality of the data

Training Time Complexity=O(n*log(n)*d)
Run-time Complexity= O (maximum depth of the tree)

- Support Vector Machine Algorithm

The Support vector machine algorithm is the classification method that solves binary classification problems. This algorithm searches for the best hyperplane that splits the data into two groups. support vector machine can obtain enigmatic relationships between data without requiring to perform a large number of transformations. Also, it can handle small and complex datasets therefore it gives better outcomes to other algorithms. The execution of the support vector machine model has been shown in Figure 5. The computational complexity of Support Vector Machine Algorithm: -
n=number of points in the Training set
k= number of Support Vectors
d=dimensionality of the data

Training Time Complexity=$O(n^2)$
Run-time Complexity= O(k*d)

- Random Forest Algorithm

The Random Forest (RF) classifier is another popular machine learning technique, which contains a large number of distinct decision trees. Each tree produces a class prediction in the RF and the class with the greatest quantity of poll takes place in the prediction of our model. It is very flexible and is capable of resolving any type of problem that can be classification or regression. The execution of the random forest model has been shown in Figure 5. The computational complexity of Random Forest Algorithm: -
n=number of points in the Training set
d=dimensionality of the data
k=number of Decision Trees

Training Time Complexity= O(n*log(n)*d*k)
Run-time Complexity= O (depth of tree* k)
Space Complexity= O (depth of tree *k)

- Gradient-Booting Tree Algorithm

The gradient-booting tree is a machine learning classification algorithm. To produce the final predictions, the gradient-booting tree amalgamates the predictions from diverse decision trees. This model is one of the most powerful techniques for building a predictive model. The execution of the gradient-booting tree model has been shown in Figure 6.
The computational complexity of Gradient-Booting Tree Algorithm: -
n=number of points in the Training set
d=dimensionality of the data
k=number of Decision Trees

Training Time Complexity= O(n*log(n)*d*k)
Run-time Complexity= O (depth of tree* k)
Space Complexity= O (depth of tree *k)

D. Evaluating the Classifiers

To assess the performance of a classification model is used a confusion matrix that is n x n (n is the number of target classes) matrix. There are compared the actual target values with the predictions of the machine learning algorithm. We have used a Confusion Matrix and Area Under ROC to evaluate the Binary classification model. The confusion matrix is used for evaluating the performance of a predictive model using the testing dataset. For evaluating the model performance, it is using variant metrics such as accuracy, precision, recall, and f-score, which are calculated using the formula provided in equations (1), (2), (3), and (4).

Table. 2 Confusion Matrix.

Predicted Value (predicted by the test)		Actual Value (as confirmed by experiment): positives	Actual Value (as confirmed by experiment): negatives
	positives	**TP** True Positive	**FP** False Positive
	negatives	**FN** False Negative	**TN** True Negative

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN} \quad \ldots (1)$$

$$Precision = \frac{TP}{TP + FP} \quad \ldots (2)$$

$$Recall = \frac{TP}{TP + FN} \quad \ldots (3)$$

Authorized licensed use limited to: University of Saskatchewan. Downloaded on August 10,2021 at 02:37:47 UTC from IEEE Xplore. Restrictions apply.

$$F1 - score = \frac{2}{\frac{1}{Recall} + \frac{1}{Precision}} \quad \ldots (4)$$

In the above equations, the number of positive samples that were correctly estimated is referred to as the true positive (TP). The number of correctly estimated negative samples is known as the true negative (TN). The number of positive samples estimated incorrectly is referred to as false negative (FN). The number of negative samples that were incorrectly predicted as positive is referred to as false positive (FP).

The area under ROC is another metric that is used for evaluating the predictive models and it is very similar to other metrics. The area under ROC shows the true positive rates against the false positive rates. The Spark MLlib includes a BinaryClassificationEvaluator class which is used to evaluate the binary classification models.

IV. Results and Discussions

A. Experimental setup

We have used a 4 core i5 processor with 16 GB of RAM and Apache Spark framework with version 2.4.7. To build a predictive model of Cardiovascular diseases, there has been used PySpark, Spark MLIib, and Dataframe libraries which are united on Apache Spark. Also, the Anaconda editor interface has been used for the execution and implementation of classification models. The execution of all models with metrics as Area Under ROC and Accuracy has been shown below.

```
#Logistic Regression Model
from pyspark.ml.classification import LogisticRegression
from pyspark.ml.evaluation import BinaryClassificationEvaluator
lr = (LogisticRegression(featuresCol='features_scaled',
labelCol="label", maxIter=10))
lrModel = lr.fit(train_data)
lrpredicted = lrModel.transform(test_data)
evaluator=BinaryClassificationEvaluator()
print('Test Area Under ROC', evaluator.evaluate(lrpredicted))
accuracy=lrpredicted.filter(lrpredicted.label==lrpredicted.prediction)
.count()/float(lrpredicted.count())
print("Accuracy :", accuracy)
y_true = lrpredicted.select(['label']).collect()
y_pred = lrpredicted.select(['prediction']).collect()
from sklearn.metrics import confusion_matrix
print(confusion_matrix(y_true, y_pred))
```

```
Test Area Under ROC 0.7633458541621477
Accuracy : 0.7071444209297234
[[5181 1716]
 [2297 4509]]
```

Figure 3. The execution of the Logistic Regression Model

```
#Decision Tree Classifier
from pyspark.ml.classification import DecisionTreeClassifier
dt = DecisionTreeClassifier(featuresCol='features_scaled',
labelCol="label", maxDepth = 3)
dtModel = dt.fit(train_data)
dtpredicted = dtModel.transform(test_data)
print('Test Area Under ROC', evaluator.evaluate(dtpredicted))
accuracy=dtpredicted.filter(dtpredicted.label==dtpredicted.prediction)
.count()/float(dtpredicted.count())
print("Accuracy :", accuracy)
y_true = dtpredicted.select(['label']).collect()
y_pred = dtpredicted.select(['prediction']).collect()
from sklearn.metrics import confusion_matrix
print(confusion_matrix(y_true, y_pred))
```

```
Test Area Under ROC 0.7291760108469822
Accuracy : 0.7246588338320076
[[5348 1549]
 [2224 4582]]
```

Figure 4. The execution of the Decision Tree Model

```
#Support Vector Machine Classifier
from pyspark.ml.classification import LinearSVC
svc = LinearSVC(maxIter=10, regParam=0.1)
lsvcModel = lsvc.fit(train_data)
lsvcpredicted = lsvcModel.transform(test_data)
print('Test Area Under ROC', evaluator.evaluate(lsvcpredicted))
accuracy=lsvcpredicted.filter(lsvcpredicted.label==lsvcpredicted.prediction)
.count()/float(lsvcpredicted.count())
print("Accuracy :", accuracy)
y_true = lsvcpredicted.select(['label']).collect()
y_pred = lsvcpredicted.select(['prediction']).collect()
from sklearn.metrics import confusion_matrix
print(confusion_matrix(y_true, y_pred))
```

```
Test Area Under ROC 0.7621312438670331
Accuracy : 0.7056848865211998
[[5437 1460]
 [2573 4233]]
```

Figure 5. The execution of the Support Vector Machine Model

```
#Random Forest Classifier
from pyspark.ml.classification import RandomForestClassifier
rf = RandomForestClassifier(featuresCol='features_scaled',
labelCol="label", numTrees=45, maxDepth=5)
rfModel = rf.fit(train_data)
rfpredicted = rfModel.transform(test_data)
print('Test Area Under ROC', evaluator.evaluate(rfpredicted))
accuracy=rfpredicted.filter(rfpredicted.label==rfpredicted.prediction)
.count()/float(rfpredicted.count())
print("Accuracy :", accuracy)
y_true = rfpredicted.select(['label']).collect()
y_pred = rfpredicted.select(['prediction']).collect()
from sklearn.metrics import confusion_matrix
print(confusion_matrix(y_true, y_pred))
```

```
Test Area Under ROC 0.7933551858800054
Accuracy : 0.7279427862511859
[[5431 1466]
 [2262 4544]]
```

Figure 6. The execution of the Random Forest Model

Authorized licensed use limited to: University of Saskatchewan. Downloaded on August 10,2021 at 02:37:47 UTC from IEEE Xplore. Restrictions apply.

```
#Gradient-Boosted Tree Classifier
from pyspark.ml.classification import GBTClassifier
gbt = GBTClassifier(labelCol='label', maxIter=75, maxDepth=3)
gbtModel = gbt.fit(train_data)
gbtpredicted = gbtModel.transform(test_data)
print('Test Area Under ROC', evaluator.evaluate(gbtpredicted))
accuracy=gbtpredicted.filter(gbtpredicted.label==gbtpredicted.prediction)
.count()/float(gbtpredicted.count())
print("Accuracy :", accuracy)
y_true = gbtpredicted.select(['label']).collect()
y_pred = gbtpredicted.select(['prediction']).collect()
from sklearn.metrics import confusion_matrix
print(confusion_matrix(y_true, y_pred))
```

```
Test Area Under ROC 0.8002401078869633
Accuracy : 0.7320294825950522
[[5237 1660]
 [2012 4794]]
```

Figure 7. The execution of the Gradient-Booting Tree Model

B. Results

Table 3. Classifiers performance

Evaluation criteria	LR	DT	SVM	RF	GBT
Accuracy(%)	70.71	72.46	70.56	72.79	73.20
ROC Score(%)	76.33	72.91	76.21	79.33	80.02
Precision(%)	72.43	74.73	74.35	75.60	76.79
Recall(%)	66.25	67.32	62.19	66.76	66.10
F1-Score	0.6920	0.7083	0.6773	0.7091	0.7105

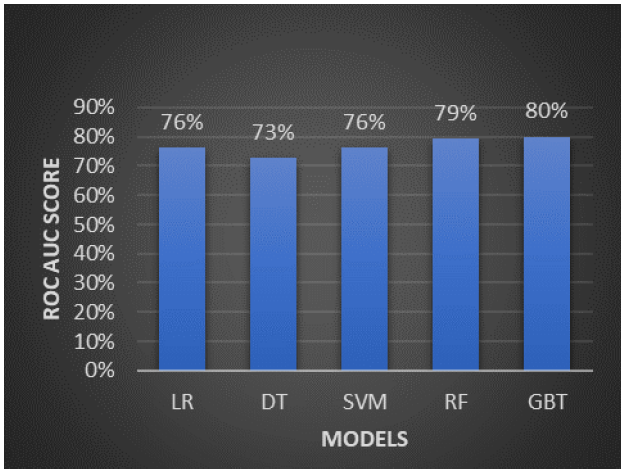


Figure 8. Comparison with Area under Roc values of all implemented models

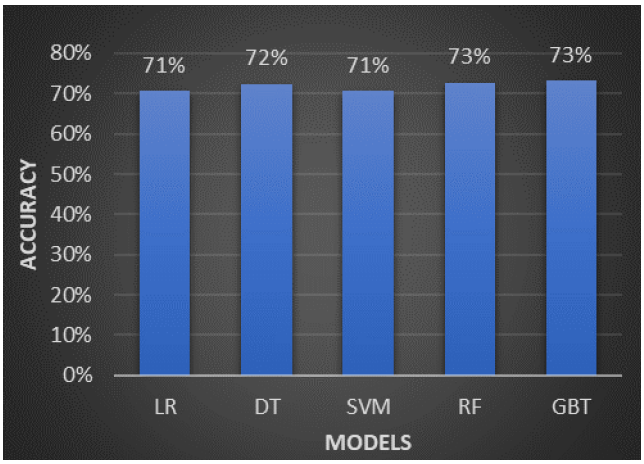


Figure 9. Comparison with accuracy values of all implemented models

In this research paper, the performance of all models is presented in Table 3. The Area Under ROC score of the gradient-boosting tree model is 0.8002 which is higher than all models. The Area Under ROC score of the random forest model is 0.7933. The logistic regression, decision tree, and support vector machine models achieved Area Under ROC scores of 0.7633, 0.7291, and 0.7621. The model that obtained higher accuracy of 73.20% than other models is the gradient-boosting tree. The decision tree and random forest model had accomplished accuracy of 72.46% and 72.79% which are almost similar. Both logistic regression and the support vector machine model had achieved an accuracy of 70.71% and 70.56%. All implemented models with metrics such as Area Under ROC and Accuracy have been shown in Figure. 8 and Figure. 9.

V. CONCLUSION AND FUTURE DIRECTIONS.

From this research work, we have built a predictive model for predicting Cardio Vascular Disease, using PySpark and Spark MLlib libraries which are integrated with the Apache Spark framework. We tried to apply supervised algorithms that included decision tree classifier, logistic regression classifier, support vector machine classifier, random forest classifier, and gradient-boosting tree classifier and for evaluating the performance of the models, have used metrics as Area Under ROC Curve and Accuracy. Included some stages for making a proposed framework such as loading cardiovascular diseases dataset, cleaning and preprocessing of the dataset, classifiers, cross-validation & hyperparameter tuning, and at the end evaluating classifiers. The gradient-boosting tree achieved the best results, and then we performed tuning with the ParamGridBuilder and the CrossValidator that help to improve the model performance. The results showed that the gradient-boosting tree classifier had accomplished the highest score of Area Under ROC and Accuracy at 80.02% and 73.20%. There will be a thorough study of huge datasets

Authorized licensed use limited to: University of Saskatchewan. Downloaded on August 10,2021 at 02:37:47 UTC from IEEE Xplore. Restrictions apply.

with more attributes to attain the highest accuracy in future work. Another future work is; researchers can use deep learning techniques to improve the performance of the model.

References

[1] Masethe, H. D., & Masethe, M. A. (2014, October). Prediction of heart disease using classification algorithms. In Proceedings of the World Congress on Engineering and computer

[2] Haq, A. U., Li, J. P., Memon, M. H., Nazir, S., & Sun, R. (2018). A hybrid intelligent system framework for the prediction of heart disease using machine learning algorithms. *Mobile Information Systems*, *2018*.

[3] Rajmohan, K., Paramasivam, I., & SathyaNarayan, S. (2014, February). Prediction and Diagnosis of Cardio Vascular Disease--A Critical Survey. In 2014 World Congress on Computing and Communication Technologies (pp. 246-251). IEEE.

[4] Sitar-tăut, A., Zdrenghea, D., Pop, D., & Sitar-tăut, D. (2009). Using machine learning algorithms in cardiovascular disease risk evaluation. Age, 1(4), 4.

[5] Mary, N., Khan, B., Ishfaq, Q., & Khan, M. Z. Empirical Study of Intelligence Techniques for Cardio Vascular Disease.

[6] Maini, E., Venkateswarlu, B., & Gupta, A. (2018, August). Applying machine learning algorithms to develop a universal cardiovascular disease prediction system. In International Conference on Intelligent Data Communication Technologies and Internet of Things (pp. 627-632). Springer, Cham.

[7] Kumar, N. K., Sindhu, G. S., Prashanthi, D. K., & Sulthana, A. S. (2020, March). Analysis and prediction of cardiovascular disease using machine learning classifiers. In 2020 6th International Conference on Advanced Computing and Communication Systems (ICACCS) (pp. 15-21). IEEE.

[8] Dewan, A., & Sharma, M. (2015, March). Prediction of heart disease using a hybrid technique in data mining classification. In 2015 2nd International Conference on Computing for Sustainable Global Development (INDIACom) (pp. 704-706). IEEE.

[9] Shamsollahi, M., Badiee, A., & Ghazanfari, M. (2019). Using combined descriptive and predictive methods of data mining for coronary artery disease prediction: a case study approach. Journal of AI and Data Mining, 7(1), 47-58.

[10] Makumba, D. O., Cheruiyot, W., & Ogada, K. (2019). A model for coronary heart disease prediction using data mining classification techniques. Asian Journal of Research in Computer Science, 1-19.

[11] Khosla, A., Cao, Y., Lin, C. C. Y., Chiu, H. K., Hu, J., & Lee, H. (2010, July). An integrated machine learning approach to stroke prediction. In Proceedings of the 16th ACM SIGKDD international conference on Knowledge discovery and data mining (pp. 183-192).

[12] Dinesh, K. G., Arumugaraj, K., Santhosh, K. D., & Mareeswari, V. (2018, March). Prediction of cardiovascular disease using machine learning algorithms. In 2018 International Conference on Current Trends towards Converging Technologies (ICCTCT) (pp. 1-7). IEEE

[13] Kaur, H., & Kumari, V. (2020). Predictive modeling and analytics for diabetes using a machine learning approach. *Applied computing and informatics*.

[14] Ahmed, H., Younis, E. M., & Ali, A. A. (2020, February). Predicting Diabetes using Distributed Machine Learning based on Apache Spark. In 2020 International Conference on Innovative Trends in Communication and Computer Engineering (ITCE) (pp. 44-49). IEEE.

[15] Ali, A. A. (2019). Stroke Prediction using Distributed Machine Learning Based on Apache Spark. *Stroke*, *28*(15), 89-97.

[16] Suma, V., & Hills, S. M. (2020). Data Mining based Prediction of Demand in Indian Market for Refurbished Electronics. *Journal of Soft Computing Paradigm (JSCP)*, *2*(03), 153-159.

[17] Kumar, T. S. (2020). Data Mining Based Marketing Decision Support System Using Hybrid Machine Learning Algorithm. *Journal of Artificial Intelligence*, *2*(03), 185-193.

[18] Anand, J. V. "A Methodology of Atmospheric Deterioration Forecasting and Evaluation through Data Mining and Business Intelligence." Journal of Ubiquitous Computing and Communication Technologies (UCCT) 2, no. 02 (2020): 79-87.

Authorized licensed use limited to: University of Saskatchewan. Downloaded on August 10,2021 at 02:37:47 UTC from IEEE Xplore. Restrictions apply.